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No. 15.—VOL. I.

SATURDAY, APRIL 12, 1884.

PRICE 1½d.

NATURE AROUND LONDON.

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by a side-hatch, in order that some large pipe might be laid down; and the big trout, with his lesser brethren, fell a victim to the predatory instincts of a party of navvies. Our author looked in vain next day into the still depths of the beech-tree pool; his finny friend was gone, and the place looked empty and dull without him.

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One particular lane, with a high hedge bordered with elm-trees, had four or five nightingales; and a copse near it resounded in the season with the cheerful call of the cuckoo. Magpies, which have become scarce in many places throughout the country, are plentiful near London, where some birds are also found which, in many country districts, are but rare and occasional visitors, such as the blackcap, shrike, and gorgeous kingfisher. To a student of bird-life, such spots as a little wood, which our author christened Nightingale Copse, cannot fail to prove a perfect paradise. It was a favourite resort not only of nightingales, but of other migratory birds—chiff-chaffs, willow wrens, golden-crested wrens, fieldfares, &c. In the fields bordering the highway, partridges abounded; and Mr Jefferies counted on one occasion as many as seventeen young pheasants all feeding

together on the wheat-stubbles. Nor is the ear the only sense which is charmed in these woodland copses—in the hedgerows, and under the straggling trees and bushes which border the woods, flowers abound, gleaming out in the sunshine from between the tall grasses with a sudden surprise of vivid colour; or spreading like enamel over the short turf; or intertwining their gay garlands with the clustering masses of creeping bramble. Each flower has its own peculiar habitat, where it flourishes luxuriantly. There are patches of the yellow rock rose, of the cranesbill, of the sweet purple wild thyme, of the starry white stitchwort, of the campion and yellow snapdragon; while stately and tall under the shadow of the birch-trees, the foxglove hangs out to the rustling breeze its lovely bells of clouded purple. Nor is heath wanting; ‘the open slopes beyond Sandown are covered with heath, growing so thickly, that even the narrow footpaths are hidden by the overhanging bushes of it. Beneath and amid the heath, what seems a species of lichen grows so profusely as to give a gray undertone to the whole.’

In autumn, this stretch of heath blazes out into a deep glory of purple, so rich and full, that it seems to give the very atmosphere a glow of purple light. Beyond the heath, there are fir-woods, stretching to the east and west; while southwards, the heath melts into the soft green of corn and meadow lands, with scattered clumps of trees. The open slopes among the straggling firs, which dot like sentinels the borders of these pine-woods, are covered with forests of tail ferns, amid which the browsing cows are lost to sight, and only reveal their whereabouts by the tinkling music of the small bells suspended to their necks.

Adders are common in these woods, and are sometimes killed for the sake of their oil, which some folks consider a specific for deafness. It is procured by skinning the adder and taking the fat and boiling it; the result being a clear oil, which never thickens even in the coldest weather. It is applied by pouring a small quantity into the ear, exactly in the same manner as the poison was poured into the ear of the sleeping king in *Hamlet*. Squirrels abound in these copses, and so do weasels and stoats.

In some fields christened by our author Magpie Fields, because he one day saw ten magpies all together in one of them, herbs abound which are in request among herbalists for medicinal purposes. One of these is yarrow. One day, looking at some mowers at work in a hayfield, he saw a man in

advance of the others pulling up the yarrow plants as fast as he could and carefully laying them aside. Asking him why he did so, he answered, that although it seemed such a common weed, it was not without its value, for that a person sometimes came and took away a whole trap-load of it. The flowers were boiled, and mixed with cayenne pepper, and were then used as a remedy for colds in the chest. Dandelions are also in request; the tender leaves are pulled in the spring, and taken away in sackfuls to be eaten as salad. There are also hellebore and blue scabious; and the rough-leaved comfrey; and borage with its reminiscences of claret-cup; and groundsel, dear to the owners of pet birds; and knotted figwort, and Aaron's rod; and a whole tribe of strongly scented mints and peppermints. The belief in these simples, which made the reputation in the middle ages of many a wonder-working doctor and village witch, is fast dying out in the country districts, where the agricultural labourers scarcely know one herb from another; but it flourishes still around the mighty and enlightened metropolis. The herb self-heal is to be found in many hedgerows of many harvest-fields, as well as on the stubbles near London; but very few reapers now would know it if they saw it, or ever think of applying it to any accidental cut or gash.

In the harvest and turnip-hoeing seasons, picturesque bivouac fires dot the fields and lanes. These do not owe their existence to parties of pleasure-seekers, who go a-gipsying under the greenwood tree, but are rather the outcome of a hard struggle for the means of subsistence. They belong to wandering Irish labourers, who move about from farm to farm wherever they can get work, sleeping in barns or outhouses, and in fine weather doing their cooking in the open air. Nothing can be more unlike the populace of the vast adjacent metropolis than these agricultural labourers, native or imported. Look at the ploughman in the furrows yonder, with his stolid characterless face, vacantly regarding the team of three stately horses before him. Intent day by day on the earth beneath his feet, he sees, or at least notices little else. 'His mind imbibes the spirit of the soil,' and cannot rise beyond. When the plough stops, he takes out his bread and cheese; and as he munches away, his eyes fall on the sunbeams glittering on the roof of the Crystal Palace; but the sparkling reflection awakens no train of thought in his uncontemplative soul; he takes no interest except in the furrows at his feet; although near London, he is not of it.

In the collection of English pottery in the Museum is preserved the simple rustic memory of these tillers of the soil, the men who, centuries ago, ploughed like this simple countryman these beautiful English acres, scattering the seed over the furrows in the green flush of spring, and garnering the golden grain beneath the mellow skies of autumn. It is curious that so much of the unwritten history of our race should be preserved by so frail a thing as earthenware. These jugs and mugs, with their quaint mottoes and ornamentation, carry the spectator back to the sports and habits of a bygone age.

‘May the best cock win,’ recalls a brutal sport now almost unknown. The frog at the bottom of the jug is a rebuke to the too greedy toper; while the motto on another cup shows that there were grumblers even in the good old days, and that times were hard then as well as now:

Here’s to thee, mine honest friend,
Wishing these hard times to mend.

Beyond the woodlands and valleys which Mr Jefferies has described so happily, are the vast South Downs, hidden in masses of gray mist. These wide sheep-walks are seemingly endless in their extent. They are profusely covered with flowers in their season, with patches of furze, and with short thick grass, amid which the wild thyme luxuriates, spreading out into soft cushions of purple which might make a seat for a king, and permeating with its aromatic fragrance the whole keen air of the uplands. The furze is full of bird-life. Only game has decreased with the increase of cultivation; and with the decrease of game, foxes have become fewer. A few years ago, they were so abundant, that a shepherd told our author that he had sometimes seen as many as six at a time sunning themselves on the precipitous face of the cliffs at Beachy Head. They ascend and descend the precipice by narrow winding-paths of their own with the greatest ease and in perfect safety, unless a couple have a quarrel on one of the narrow rock-ledges, when fatal results often ensue—one or both toppling over.

‘Lands of gold,’ says our author, ‘have been found, and lands of spices and precious merchandise; but the South Downs are the land of health. There is always the delicious air, turn where you will; and the grass, the very touch of which refreshes.’ Besides all this, there is the peculiar beauty which gives

its chief charm to all elevated situations, the interest of the panorama which spreads around and beneath—the distant trees which wave in the freshening breeze; the gleam of light which brings out into strong relief the warm bit of colouring supplied by the tiled roof of yonder farmhouse; the flashes of sunshine which brighten up the gloom, and chase the shadows across the swelling uplands and green low-lying meadows beyond.

Seen in the shifting lights and glooms of a breezy autumn day, this lofty, lonely spot seems a land of enchanted beauty, which holds the spectator spellbound, till masses of cloud, rolling up from the sea, throw deep purple shadows over the peaceful landscape, and warn him that darkness is about to fall over the flower-spangled slopes and gleaming sea beyond.

BY MEAD AND STREAM.

CHAPTER XXIII.—CHANGES.

THE arrival of a stranger in Kingshope was not such an unusual occurrence as to attract much particular attention. The villagers were accustomed in the summer to frequent visits of bands of 'beanfeasters' or 'wayzegoose' parties, as the annual outings of the employees of large city firms are called. On these occasions there were athletic games on the common, pleasant roamings through the Forest, and high revel in the *King's Head* or the *Cherry Tree* afterwards. Then there were itinerant photographers, negro minstrels, and gypsy cheap Jacks, with caravans drawn by animals which may be best described as the skeletons of horses in skin-tights—working the Forest 'pitch' or 'lay'—these being the slang terms for any given scene of operation for the professional vagrant. The bird-snarers and the pigeon-flyers seemed to be always about. In the hunting season there were generally a few guests at the *King's Head*; and so, although every new visitor underwent a bovine stare, he was forgotten as soon as he passed out of sight.

Mr Beecham's ways were so quiet, that before he had been a week in the place, he had glided so imperceptibly into its ordinary life that he seemed to be as much a part of it as the parson and the doctor. His presence was of course observed, but there was little sign of impertinent curiosity. It was understood that he was looking about the district for a suitable house in which to settle, or for a site on which to build one. This accounted for his long walks; and there was nothing remarkable in the fact that his peregrinations led him frequently by Willowmere, and sometimes into the neighbourhood of Ringsford Manor.

Although his ways were so quiet, there was nothing reserved or mysterious about them. The object which had brought him to Kingshope was easily comprehended; he entered into conversation with the people he met, and took an interest in the affairs of the place—the crops, the weather, and the prospects of the poor during the coming winter. Yet nothing more was

known of his antecedents than that he came from London, and that he visited the city two or three times a week. He dressed plainly; he lived moderately at the inn—not like one who required to reckon his expenses carefully, but like one whose tastes were simple and easily satisfied.

The general belief was that he had belonged to one of the professions, and that he had retired on a moderate competence, in order to devote his time to study of some sort. He himself said nothing on the subject.

One of the first acquaintances he made was Uncle Dick, who adhered to the kindly old country custom of giving the time of day to any one he met in the lanes or saw passing his gates. The first salutation of the master of Willowmere induced Mr Beecham to make inquiries about the district, which led to future conversations. These would have speedily introduced the stranger to the farmhouse and its mistress; but hitherto he had not availed himself of the cordial invitation which was given him. He was apparently satisfied with the privilege of going over the land with Uncle Dick, inspecting his stock and admiring his horses, and thus speedily developing a casual acquaintanceship into a friendship. On these occasions he had opportunities of seeing and conversing with Madge, and she formed as favourable an opinion of him as her uncle had done.

‘Has he ever said what made him think of coming to settle hereabout?’ inquired the dame one day, after listening to their praises of the stranger.

‘Never thought of asking him,’ replied Crawshay, wondering if there was anything wrong in having neglected to put such a natural question.

‘He mentioned that some friends of his lived near here at one time,’ said Madge, ‘and that he had always liked the Forest.’

‘Has he spoken about any family? Is he married? Has he any children?’

‘Why, mother, you wouldn’t have me go prying into what doesn’t concern us!’ was Crawshay’s exclamation. ‘It does seem a bit queer, though, that he seems to have nobody belonging to him.’

Aunt Hessy thought it very queer; and when Philip came next, she asked him to describe Mr Shield to her again.

‘He must have changed very much since I last saw him,’ she said thoughtfully. ‘I scarcely know what put it into my head, but this Mr

Beecham is much more like what I should have fancied your uncle would grow into, than the gentleman you describe. But foreign parts do seem to alter people strangely. There was neighbour Hartopp's lad went away to California; and when he came back ten years after, it took his own mother two whole days before she would believe that he was himself. Yes, foreign parts do alter people strangely in appearances as well as feelings.'

It was regarded by the little group as a good joke that Aunt Hessy should have formed the romantic suspicion that the stranger in the village might be her old friend Austin Shield. They did not know anything of the confidential letter. She had said nothing about it yet, and her conscience was much troubled on that account.

'It's wrong to keep a secret from Dick,' she kept saying to herself. 'I know it is wrong, and I am doing it. If harm come of it, I shall never forgive myself; I hope others may be able to do it.'

She regarded with something like fear the enthusiasm with which Philip spoke of the social revolution he was to effect by means of the wealth placed at his command. Yet it was a noble object the youth was aiming at. Surely wealth could do no harm, when it was used for the purpose of making the miserable happy, of showing men how they might prosper, and teaching them the great lesson, that content and comfort were only to be found in hard work. The scheme looked so feasible to her, and was so good, that she remained silent lest she should mar the work. She bore the stings of conscience, and prayed that Philip might pass safely through the ordeal to which he was unconsciously being subjected. He talked of the bounty of his uncle, and she was uneasy, knowing that this bounty might prove his ruin, although she was quite unable to see how that could come about as matters looked at present. She was simply afraid, and began to understand why preachers often spoke of gold as a fiend—the more dangerous because it appeared as the agent of good. Then there was the coming of this stranger at the same time that Philip met his uncle in London. Of course there was nothing to associate the two in her mind except the period of their arrival. But she was puzzled.

'There is not the slightest resemblance between the two men, I assure you,' Philip said; 'but there is this strong resemblance between my uncle as he is now and as he was, by your own account, when you knew him long ago—

he is as odd in his ways as ever. He will not discuss anything with me except by letter. That, you might say, was no more than prudent, as it can leave no room for dispute as to what we say to each other.'

'He wants to make you careful,' said the dame, with some feeling of relief; for this arrangement seemed to prove that he was desirous of helping Philip to pass the test.

'But, besides, he will scarcely see me at all; and when he does, he is as short with me and in as great a hurry to get rid of me as he was on the first day I called on him. When I try to explain things to him, he says: "All right; go your own way. If you want me to consider anything, you must write it out for me." I don't mind it now, having got used to it; but sometimes I cannot help wondering'——

Philip checked himself, as if he had been about to say something which he suddenly remembered should not be spoken even to his dearest friends.

'Well?' queried Uncle Dick, looking at him along the line of his churchwarden pipe as if it were a gun and he were taking aim. 'What are you stopping for? You can't help wondering at what?'

'Only at his droll ways,' answered Philip. 'I should have thought that risking so much money in my hand, he would have been anxious to have the fullest particulars of all that I was doing with it.'

'So should I, lad. What does your father say about it?'

'Nothing more than that he will want to speak to me one day soon. He is not pleased.'

'There don't seem to me much to be not pleased about.—Eh, mother?'

'We'll see after a bit,' answered the dame, cautiously, but smiling. 'We don't know yet whether Philip is to prove himself a very wise man or'——

'Or a fool,' interrupted Crawshay, with one of his hearty laughs.

'Nay, Dick; not that. Philip will never prove himself a fool; but he might do worse—he might prove himself a sensible man doing foolish things.'

The stranger who provoked this discussion went on in his calm way, seeking what apparently he could not find, but always with a pleasant smile or a kindly 'good-day' to the people he met in the fields and lanes.

One of his favourite halting-places was at the stile which gave access from the roadway to the Willowmere meadows. On the opposite side of the road were the willows and beeches, bordering the river. Four of the latter trees were known as the 'dancing beeches,' from the position in which they stood, as if they had suddenly halted whilst whirling round in a country-dance; and when the wind blew, their branches interlaced and creaked in unison, as if they wanted to begin the dance again. This was a famous trysting-place, and in the summer-time the swains and their maidens would 'wander in the meadows where the May-flowers grow.' This is the burden of a rustic ballad which you would often hear chanted in the quiet evenings. It served the double purpose of supplying the place of conversation and of agreeably expressing the thoughts of the singers. Uncle Dick sometimes saw and heard them; but with kindly indifference to his clover, he would shake his head and turn away, remembering that he, too, had once been young.

Mr Beecham resting on the stile could, by an easy movement of the head, command nearly the whole of the hollow in which the village lay; and looking upward, could catch glimpses of Willowmere House peering through the apple and pear trees of the orchard.

After the lapse of years, how new it all looks, and yet how old; how changed, and yet familiar. There is the church, the same gray weather-beaten pile, in spite of the vicar's manful efforts to get it put into a state of thorough repair. The vicar himself is the same cheery good friend in gladness, and the sympathising comforter in sorrow; his hair is almost gray now, and his figure is inclined to be rotund; but he is still the same. There are, however, new gravestones in the churchyard, and they bear the names of old friends. Their places in the world have been easily filled up; their places in the memory of the survivors never can be. Ay, there is change indeed.

But here is the golden autumn, its lustre slowly growing dim under the touch of approaching winter; there are the green fields and the red ploughed lands—they are just as they looked long ago, although his eyes see them through the sad haze which separates him from the past. There are the sounds of the cattle, the ripple of the river, and the rustle of the trees—sounds to which he gave no particular heed in the old time, and now they are like the voices of welcoming friends.

So the present steps by us; pain and sorrow plant milestones on our way; by-and-by the eye glances tenderly backward and over them, and in old age we hear the voices of our youth.

‘Good-afternoon, Mr Beecham. Do you think it will rain?’

He lifted his head, and bowed to Madge and Philip as they were about to pass over the stile. He looked up at the sky.

‘I am afraid it will rain; but you will be home before it begins, I think.’

Philip gave her his hand; she mounted the three foot-worn wooden steps and descended on the meadow side.

‘I hope you will always have a strong hand to help you over the stiles, Miss Heathcote,’ he said, smiling; but there seemed to be as much of earnest as of jest in his meaning.

‘I believe she may fairly count upon that, Mr Beecham,’ answered Philip.

‘The pity is, we so seldom find what we count upon,’ said Mr Beecham, shaking his head.

‘Then we must make the best of what we do find,’ replied Philip cheerfully, ‘and scramble over somehow without a helping hand.’

The two passed on at a smart pace up the meadow, Mr Beecham looking after them with a dream in his eyes.

Overhead, on this afternoon, was a sky gloomy and threatening; but on the horizon were rivers of pale golden light, giving hope and courage to the weary ones who were like to faint by the wayside. Suddenly a white light relieved the gloom immediately above, and the golden rivers were lashed with dark promontories; but still, the farthest point was light. Again suddenly a white glory burst through the gloom, dazzling the eyes and breaking the clouds into fantastic shapes, which fled from it like the witches of evil fleeing before the majestic genii of good. Another change, and all gradually toned down into the soft repose of a calm evening, bearing the promise of a pleasant day to follow.

‘I have lived alone too much,’ muttered Mr Beecham with a long-drawn breath, which is the only approach to a sigh ventured upon by a man past middle age; ‘and my own morbid broodings make me superstitious,

showing me symbols in everything. I hope this one may turn out well, however.'

Philip and Madge had disappeared by this time, and Mr Beecham walked slowly on to the village.

When the young people reached the homestead, Madge announced that Philip had come to tell them something very important, which he had refused to reveal until they should be in the house.

Aunt Hessy glanced uneasily from one to the other; but seeing no sign of disturbance on either face, her uneasiness passed away. She concluded that it was some jest with which Philip had been teasing Madge.

'I have seen Mr Shield again to-day,' he began, 'and I have received new instructions from him.'

'He is not going to send you off to Griqualand, after all?' queried Madge quickly.

'O no; but maybe you would prefer that he should order me off there, rather than tell me to take chambers in town.'

'Chambers in town! What can that be for?'

'Well, he was as short and bustling as ever; he never seems to have time to discuss anything. "That's what I want," he says; "if you don't like it, write, and tell me why." All he said about it was that he desired me to feel independent.'

The uneasy expression reappeared on Aunt Hessy's face.

'Have you consented to make this change?' she asked quietly.

'I could see no objection; and in several ways the arrangement will be convenient. I made it clear that it was not in any way to be considered as a step towards separating me from my family. He said I could please myself as regarded my family—he had nothing to do with that.... Do you not like it, Madge?'

The clear eyes looked wistfully in his face. 'No, Philip; I do not like it. But perhaps Mr Shield is right; and it may be as well that you should have the experience of being away from us for a time at least.'

‘Living away from you! Why I shall be here as often as ever!’

She said nothing; and Aunt Hessy put the apparently irrelevant question:

‘Have you seen Mr Beecham to-day, Madge?’

‘We saw him by the stile at the foot of the meadow as we passed.’

Aunt Hessy, with evident disappointment, abandoned the droll fancy which had for a time possessed her mind.

SOME QUEER DISHES.

IF, in England, a man was pushed to discover a new animal food, it would, I think, be a long time before he hit upon bats as at all likely to furnish him with a desirable addition to his table, even if their diminutive size did not place an insuperable obstacle in the way of their being so utilised. But in many of the South Sea Islands where the flying-fox—a species of bat, fifteen inches or so across the wings—is common, it is used as food by the natives, and its flesh is by no means to be despised even by epicures. This animal, frugivorous in his tastes as a rule, does not for all that turn up his nose at a plump moth or a succulent beetle when they chance to come in his way; but he usually confines himself to fruit—ripe bananas of the best quality and plenty of them being about his mark; and dreadful havoc he and his friends would make in the banana gardens, if the natives—well aware of his habits—did not hasten to bind quantities of dead leaves round the ripening fruit, and so preserve it from his attacks. It would seem absurd to a stranger to the country to be informed that such an insignificant animal as a bat could seriously threaten the fruit-harvest in countries where it is so abundant; but he would change his opinion when informed that the flying-foxes often settle in hundreds in any likely plantation; and as they always destroy very much more than they consume, the loss and inconvenience they cause to the natives may be properly estimated.

The bat in question is not so strictly nocturnal in his habits as his English brother; and although he usually sallies out at sunset, yet I have often noticed them sailing about in broad daylight, provided the weather was dull and overcast; the flight is even and regular, very like that of a rook, and not in the least resembling the extremely erratic mode of progression affected by our native species. If in their manner of flying—a few steady flaps and then a long sail—they remind one of the rook, they also resemble our old friend in their habit of assembling together at bedtime, when they all retire to roost on the same grove of trees, and hang head downwards with their wings wrapped round their bodies, looking like a collection of large cobwebs.

It must not, however, be supposed that the meeting and subsequent proceedings take place in silence; the contrary is the case; and an immense amount of chattering is carried on for a considerable time, when no doubt all the affairs of the day are duly discussed, as well as other matters amatory and otherwise. In the old heathen times, the rookeries were strongly tabooed by the priests; and even to the present day, the natives, more especially the old men, have an evident aversion to interfere with the sacred trees, a feeling which does not in the least prevent them from killing all the bats they can in other places.

The natives prepare them for food by first cutting off the wings and then passing the body through the fire, to remove the fur, and with it the strong foxy smell with which it is impregnated. It is then carefully scraped, split open, and afterwards grilled on the coals spitchcock fashion, when it is ready for consumption; and is capital eating, having a rich gamy flavour something between a hare and a woodcock.

I was so much encouraged by the success of my first essay at bat-eating, that I afterwards had a pie made of several I had shot, and from my previous experience, rather looked forward to a good dinner; but when the pastry was cut open, I was grievously disappointed by finding that the fetid odour peculiar to the live animal had survived the cooking—from being unable to escape from the pastry—rendering it utterly uneatable, and so for the future contented myself with bat *au naturel*—that is, native fashion.

The above-mentioned animal is very common in Australia, and is quite as great a nuisance among the orchards there as he is in the islands; but it will be some considerable time, I fancy, before our colonial brothers utilise him in the kitchen.

I don't suppose that many people—at least English people, who are tolerably prejudiced in their way—have ever voluntarily gone in for a cuttle-fish or octopus diet, as they are horribly weird, uncanny animals to look at; and few, I opine, would feel inclined to make a 'square meal' off the shiny creatures, at least until other more prepossessing kinds of food remained to be tried. Nevertheless, throughout the whole of the Pacific, including Japan, all the different varieties of cuttle and octopus are regarded as a *bonne bouche* of peculiar excellence; and both in its capture and preparation, the natives display considerable ingenuity. I remember once,

when sailing in the tropics, seeing one morning the deck of our little schooner nearly covered with that very elegant little cuttle-fish called the 'flying-squid.' The sea had been very rough during the night, and I could never properly ascertain whether the squid had come on board of their own accord, attracted by the light—as the men affirmed—or had been left there by a heavy sea we had shipped just before daylight. Anyway, our cook, a smart Maltese, at once set to work to collect them, and then, much to the disgust of the sailors, who are the most prejudiced of mortals, he forthwith proceeded to cook them for the cabin table, and sent us down dishes of squid both curried and fried that were much approved of by all who partook of them; and proved a delightful change after the long course of 'salt junk' and tinned soup and bouillie that the slow sailing of our little craft had obliged us to adopt.

These fish were about six inches long, had large brilliant eyes of a set expression, and were furnished with a pair of flippers or wings. They also—unlike any other kind of fish that I am acquainted with—rejoice in a couple of tails, in lieu of the orthodox number. The body, almost transparent, was of a delicate olive brown. Altogether, they were pretty little things, and tasted even better than they looked.

I am now about to introduce my readers to a dish of octopus prepared *secundum artem* by a South Sea native. The octopus is by no means, without proper apparatus, an easy animal to lay hold of; on the contrary, it demands all the cunning of the most experienced South Sea fisherman to wile him from his haunts in the coral and to secure a good number for a feast.

But here is my Tongese friend Fakatene, just about to launch his *hamatefna*, or fishing-canoe; and we cannot do better than accompany him on his trip, and lend a hand in catching the fish we are to partake of. But first, just notice how ingeniously his tiny vessel is constructed out of timber of the bread-fruit tree. This tree does not, so far south—we are in about twenty-three degrees five minutes south—attain to any great size, and the timber, therefore, is proportionately small and scarce, which accounts for the small size of the pieces used. The hull, you notice, is pretty well in one piece, except that queer-shaped bit so artfully let in near the bows, and so close-fitting all round that even a penknife could not be introduced between the seams; and were it not for the difference in the grain of the wood, the

ingenious patch would never be detected. The top sides are formed of several small planks neatly sewn on to the hull with sinnet, and joined in the same manner to one another; and yet, with all this patching, she exceeds in beauty, in the grace of her lines, and in her extreme buoyancy in the water, the finest four-oar ever turned out by Searle in his most palmy days.

Fakatene is pleased with our admiration of his highly prized canoe, and takes some pains to explain that she was moulded on the lines of the bonito, one of the swiftest of fishes. Not such a bad idea that, we consider, for a poor native; but one that we intellectual white men are much too proud, not to say too conceited, to follow; so we go in for all kinds of scientific curves and angles, with the result that our builders are constantly producing craft that will neither pull nor sail, and that would have been a disgrace to Noah himself, or even to prehistoric man.—But to return to our canoe. She is provided with an outrigger called a ‘thama,’ to prevent capsizing; with a carved-wood bailer, in case we ship a sea or make any water from the working of the seams; also with a long three-pronged fish-spear, a few lines, a bamboo of fresh water; and last, but not least, with the inevitable fire-stick, or smouldering twist of tapa cloth, to furnish a light for our friend’s *seluka* (cigarette). Off at last; and Fakatene, who poled swiftly over the shallow part of the reef, has taken to his paddle, and coasting along the island for some distance, we soon come to a favourable spot for our purpose; so we drop anchor—a large stone—and business commences.

The octopus dwells in holes in the reef, keeping only a portion of his body exposed, so that, while he can look out for his prey, he can at the same time quickly withdraw within his hole, directly his dread enemy the shark appears, who is always foraging about the reefs in search of adventurous cuttles.

Now, I must tell you that the octopus, although partial enough to crabs, is particularly fond of the inhabitant of the spotted cowrie or ear-shell, so common in our shops; and so Fakatene, well aware of this fact, has prepared a cunning bait, artfully constructed of a number of small plates of the shell fastened together in such a manner that while similar in appearance to the real thing, yet, being much heavier, and not containing any air, sinks at once, which a real shell would not do. Our friend now lowers his line, with the shell-bait attached, until it touches the bottom, and then raising it a few inches off the ground, jerks it gently up and down.

Presently, a pull on the line shows that the fish has taken the bait; more jerking on the part of the native; which the octopus replies to by at once throwing out a fresh arm. The jerking still continues; until the fish, dreading the escape of his prey, lets go his hold of the rocks, and wraps the whole of his body round the shell; when the native, perceiving that his line is no longer fast to the ground, gently hauls up the line, and finally deposits an immense octopus in the bottom of the canoe. Our new friend no sooner finds himself caught, than he lets go the deceptive bait, and with his great goggle eyes staring hard at nothing in particular, sprawls about in the most awkward fashion, at the same time giving vent to a species of grunt, until at last he finally retires into the darkest corner he can find, and collapses into a lump of grayish-looking jelly, about a third part of his apparent size when in motion.

Having by the same means secured several more fish, we return to land, when the canoe is duly housed, and Fakatene disposes of the octopi by turning them inside out and hanging them up to dry in the sun, having first carefully saved all the sepia left in the fish, as this is esteemed a great luxury, and an indispensable ingredient in preparing the sauce.

When the cuttle is to be cooked, it is first of all carefully cleaned and scraped, when all the outer skin, including the hideous-looking suckers, comes off. The fish is then cut in pieces, and having been tied up in a banana-leaf, is baked in an oven for a considerable time in conjunction with cocoa-nut milk and a certain proportion of the inky-hued sepia above mentioned, and which, as is well known, is made use of by the fish when alive to obscure the water when escaping from the pursuit of its enemies. It takes some time to cook octopus properly, as it is naturally tough and stringy; but when well prepared, it is one of the most delicate and luscious dishes I ever tasted; and, singular to say, the cooking converts the tripy, stringy-looking substance into a solid meaty food, bearing a curious resemblance to lobster both in taste and colour, only rather firmer in texture; a most unlooked-for occurrence in such dissimilar articles.

A WITNESS FOR THE DEFENCE.

IN THREE CHAPTERS.—CHAPTER II.

WHEN I got back to town, the sessions were only a week off; so the first thing I did was to call on the solicitor in charge of my murder case, in order to learn from him how it stood, and to take it off his hands. The magistrate, of course, had sent the prisoner for trial. When I came to read the depositions, the case against him seemed perfectly simple, and as conclusive as circumstantial evidence could make it. The crime had not occurred so long ago but that a diligent search had unearthed several witnesses. The servant-girl, who had now become the wife of a dairyman in the immediate neighbourhood, was found. She proved the bad conduct of young Harden, and the ill-will which gradually grew up between him and her former mistress. She also spoke to his ejection from the house on the day of the murder, and to his threats at the street-door. She swore to the knife, which had been in the possession of the police ever since, as having belonged to the prisoner. There were other witnesses to the same facts; and the landlord, my client, and several others, proved the flourishing of the identical knife and the ominous words in the public-house. To complete the chain, the man who had instructed me proved the finding of the knife in the room where the murder was committed; and two or three witnesses remembered being by his side and seeing him stoop down and pick it up. These, with the final facts of his sudden disappearance and changes of name, appeared both to me and to my friend to be capable of being spun into a rope quite strong enough to swing John Harden out of the world.

‘But,’ said my solicitor-friend, ‘the queerest thing of all is that no one is going to appear for the prisoner.’

‘No one to appear for him?’

‘No one. Young Elkin holds a watching brief on behalf of the prisoner’s master, and that is all. He said Harden had been in Mr Slocum’s—that’s his master—service for over seven years, behaving extremely well all the time.

He was invaluable to his old master, who is something of an invalid. He had turned religious, and was disgusted at his former wicked life.'

'But I suppose he has money—or, at anyrate, if Slocum is so fond of him, why doesn't he pay for the defence?'

'Why, it seems that his notion of religion forbids Harden to avail himself of worldly arts. Slocum is only too anxious to retain some one; but Harden won't have it, and no one can persuade him. Says he is in the hands of a Higher Power, and it shall be given him what he shall speak, and all the rest of it. He wanted to make a speech to the magistrate; but Slocum, by Elkin's advice, did manage to induce him to hold his tongue for the present, and say he would reserve his defence. Of course they hope he will come to his senses before the trial. But I don't know how that will be. I never saw such an obstinate pig. Only gave in to his master about not speaking because the poor man began to whimper in court!'

The main part of my work had been done for me, and it only remained to bespeak copies of the depositions, see the witnesses, and make sure that they intended to say at the Old Bailey substantially the same things as they had said at the police court—a most necessary precaution, the imagination being so vivid in people of this class that they are very likely to amplify their tale if possible—and prepare the brief for the prosecuting counsel. This done, I had but to let things take their course.

When the day of the trial came, I was betimes in my place at the Central Criminal Court, having various other cases in hand there. The prisoners, as is customary, were first put up and arraigned—that is, had the substance of their several indictments read over to them—and were called on to plead 'guilty' or 'not guilty.' These disposed of, the case for John Harden was called, and I looked at him with some curiosity. No sooner had I done so than I knew that his was a face upon which at some time or other I had looked before, and of which I had taken note. It is a useful peculiarity of mine that I never forget a face to which I have once paid any attention, and I can generally recollect the place and circumstances under which I last saw it. But here the latter part of my powers failed me. I knew the face well, but could not imagine when and where I had beheld it. I even knew that I had seen the man bare-headed, and that he was not then, as now, bald on the

crown. The thing worried me not a little. In the meanwhile, John Harden was being put up to take his trial for the murder of Agatha Harden.

‘I, m’lud, appear to prosecute in this case,’ said my counsel, starting up and down again like the blade of a knife.

‘Does nobody appear for the prisoner?’ asked the judge.

‘I understand, m’lud, that the prisoner is not represented,’ said counsel, appearing and disappearing as before.

‘My lord,’ said an agitated voice from the body of the court, ‘I have used all possible efforts’——

‘Si-lence!’ proclaimed the usher.

‘Who is that?’ inquired the judge, looking over his spectacles.

‘My lord, I am this foolish fellow’s master; and I am perfectly convinced’——

‘I cannot hear you, sir. If the prisoner wishes to have counsel assigned to him for his defence, I will name a gentleman, and will take care that the prisoner shall have due opportunity for his instruction; and if you desire to give evidence on his behalf, you can do so.—Prisoner, is it your wish that counsel be assigned to you for your defence?’

Harden had been standing with his head slightly bent, and his clasped hands resting on the rail of the dock. He now looked up at the judge, and replied in a grave and impassive voice: ‘My lord, I wish no help but the help of God. I am in His hands, and I am an innocent man. If He sees good to deliver me, He will do so. Who am I, that I should interfere with His work?’

‘You appear to me,’ said the judge gently, ‘to be under an unfortunate delusion. You say rightly that you are in God’s hands; but that should not hinder you from using such instruments for your deliverance as he offers you. Once more I will ask, do you now desire to be represented by counsel?’

‘I do not, my lord.’

‘So be it.—Now, Mr Clincher.’

Rising once more, counsel for the prosecution proceeded to open his case. It was clear and straightforward, put concisely and tellingly, and embraced the facts which the reader already knows. He then called his witnesses; and as each after each left the box, it was easy to see from the faces of the jury that things were likely to go hard with the prisoner. Always, in answer to the inquiry, 'Do you wish to put any questions to this witness?' Harden replied: 'No, my lord. He has said the truth, for all I know.'

So smoothly did the trial run its course, that only one incident called for remark. This was when my client got into the box; and so indecently eager did he appear to be to procure the conviction of the prisoner, that he twice called down upon himself a severe rebuke from the judge, for persistently volunteering irrelevant statements to Harden's prejudice. And when counsel at length said, 'That, m'lud, is my case,' and sat down, but little doubt remained as to the prisoner's fate. I still sat with my gaze fascinated by the set face in the dock, trying—trying to remember when and where I had last looked upon it.

'Do you propose, prisoner, to call any witnesses?' asked the judge.

'Only my master, my lord—Mr Slocum. He'll speak for me, and he'll say, I know, that I'm not the man to kill any living thing.'

'Very well.—And now, before calling him, do you desire to address the jury?'

The interest of the case, which, except for that interest which is inseparable from a trial for murder, had slightly flagged, revived now that a human being was virtually at grips with death. For what had just passed meant that there was no defence or attempt at a defence, that the jury must convict, and that the man must die, without hope of mercy for so cowardly and ungrateful a murderer. There was not a sound in the court. It was late in the afternoon, and the winter sun was setting. Its rays lit up the crimson hangings, the scarlet robes of the judge, the intent faces, all looking one way, the drooping head and white composed countenance of the prisoner—the man standing up there in full health and strength, and whose life was going down with the sun.

'I have but a few words to say, my lord and gentlemen. I didn't do it. I was bad enough, and maybe cruel enough in those days, to do it; but I didn't. I

was so drunk and so mad, my lord and gentlemen, that I might have done it if it had happened earlier in the day, unknown almost to myself, and be standing here rightly enough. But I *know* I couldn't have done it, and why? Because I was miles away at the time. My poor aunt, as I've heard from what has been said, must have been killed between a quarter to and a quarter past eight in the evening. Well, at eight o'clock I was at least five miles off. If I'd done it directly the girl went out of the house—as she says, at a quarter to eight—it isn't according to reason that I could have broke open the cupboard, took the money, and got five miles off in a quarter of an hour.' He stopped, and drew the cuff of his coat across his forehead.

Where *had* I seen him before? Where and when had I seen him do that very action?

'O gentlemen, I couldn't have done it! I couldn't, bad as I was! I know, now, how bad that must have been—the mercy of God has been upon me since those days—but bad as I was, I owed her too much, and knew it, to have hurt her in any way. Won't you believe me? I tell you I was miles away at the time—miles away. Who can tell us you're saying true? you will ask. No one, I suppose. Not a soul was near me that I knew, to come here and speak the truth for me this day. But I know the same God that saved Daniel can save me from a sorry end, if it is His will to do it—if not, His will be done! I'm keeping you too long, only saying the same over and over again. I'll just tell you how it was, and I've done, and you must do as duty bids you.'

Another pause. The silence of death, or rather of a deathbed. The faces in the distance of the darkened court shimmered through the gloom, like those of spectres waiting to welcome a coming shade. Then the gas-light burst forth, and all sprang into sudden distinctness, and there was a general half-stir as of relief.

'Oh, isn't there one here that can speak for me? Is there any one who remembers the great gas-main explosion in —— Street that year?'

There was again a stir, and a more decided one. Clearly there were many in court who remembered it. I did, for one. And remembering it, I seemed as one in a tunnel, who sees the glimmer from the distant opening, but can distinguish no feature of the landscape beyond.

‘I was there—that night. It was the night of the day I was turned out of doors—the night of the murder. How I came to be there, so far from my aunt’s neighbourhood, I don’t know, but I found myself working hard, helping to lift the stones and timber of the house-fronts that were blown in, and getting the poor crushed people out. I worked a long time, till I was like to drop; and a policeman clapped me on the back and gave me a word of praise and a drink of beer out of a can. I wonder where that policeman is now, and if he’d remember?’

He did not respond, wherever he might be. No one to help—no friendly plank to bridge over the yawning grave. What was it, this that I was trying so hard to recall?

‘I wandered off after that into the by-streets. I knew those parts well. I had had a comrade who used to live there, and many a wicked and foolish prank we’d played thereabouts. The beer I had just drunk on an empty stomach had muddled me again a bit, but I was quite sober enough to know every step of the way I went, and remember it now. I turned up Hoadley Street, and then to the left along Blewitt Street; and just when my aunt must have been struggling with the wretch that took her life, whoever it was, I heard a clock strike eight. I did, gentlemen, and I suppose I never thought of it since; but now I remember it as clear as day. I was standing at the time at the corner of Hauraki Street.’

It all came back to me in a moment! I heard the patter of the rain on the cab-roof—I saw the gleam of the infrequent lights on the wet flags—I listened to the objurgations of the cabman at the obstructing dray—I took note of the reflection in the mirror, the queer street-name which would not rhyme so as to make sense. The strokes of the clock striking eight were in my ears. I saw the lamp at the corner, and the man underneath looking up at it—the man with the short broad face, the sharp chin, the long thin mouth turned down at the corners, and the blank in the front teeth—the innocent man I was hounding to his death—the prisoner at the bar!

As I sprang to my feet, down with a crash went my bag full of papers, my hat and umbrella, so that even the impassive judge gave a start, and the usher, waking up, once more proclaimed ‘Si-lence!’ with shocked and injured inflection. Heedless of the majesty of the law, I beckoned to my counsel, and as he leaned over to me in surprise, I whispered earnestly in

his ear. I never saw the human face express more entire astonishment. However, seeing that I was unmistakably in earnest, he merely nodded and rose to his feet.

‘Your lordship will pardon me,’ he said, ‘for interfering at this stage between the prisoner and the jury; but I am instructed to make a communication which I feel sure will be as astounding to your lordship and the jury as it is to myself. I think I may say that it is the most surprising and unprecedented thing which ever occurred in a court of justice. My lord, the solicitor who instructs me to prosecute tenders himself as a witness for the defence!’

OUR HEALTH.

BY DR ANDREW WILSON, F.R.S.E.

II. FOOD AND HEALTH.

FROM the point of view of the political economist, the idle man has no right to participate in the food-supply of the active worker. Whatever may be the correctness and force of the arguments which the economist may use by way of proving that the non-worker and non-producer has no right to participate in the ordinary nutritive supply of his fellows, the physiological standpoint assumes another and different aspect. The idle man grows hungry and thirsty with the regularity of the man who works. He demands food and drink as does his energetic companion; and the plea that idleness can need no food-support, may be met in a singularly happy and forcible fashion by a plain scientific consideration. In the first instance, the idle man might, by an appeal to science, show, that whilst he apparently spent life without exertion, his bodily functions really represented in their ordinary working an immense amount of labour. Sleeping or waking, that bodily pumping-engine the heart does not fail to discharge its work, in the circulation of the blood. The rise and fall of the chest in the sleeping man remind us that it is not death but his 'twin-brother sleep,' that we are observing. If we make a calculation respecting the work which the heart of a man, idle or active, performs in twenty-four hours, we may discover that it represents an amount of labour equal to one hundred and twenty foot-tons. That is to say, if we could gather all the force expended by the heart during its work of twenty-four hours into one huge lift, such force would be equal to that required to raise one hundred and twenty tons-weight one foot high. Similarly, the work of the muscles of breathing in twenty-four hours, represents a force equal to that required to lift twenty-one tons one foot high. These are only two examples out of many, which the ordinary work and labour of mere vegetative existence, without taking into consideration any work performed—in the popular sense of the term—involves.

We thus discover that, apart altogether from the every-day labour of life, in which brain and muscles engage, an immense amount of work is performed in the mere act of keeping ourselves alive. Nowhere in nature is work performed without proportionate waste, or wear and tear of the machine that works. This dictum holds quite as true of the human body as of the steam-engine. And as the engine or other machine requires to be supplied with the conditions necessary for the production of force, so the living body similarly demands a supply of material from which its energy (or the power of doing work) can be derived. As the engine obtains the necessary conditions from the fuel and water it consumes, so the living body derives its energy from the food upon which it subsists. Food in this light is therefore merely matter taken from the outside world, and from which our bodies derive the substances required for the repair of the waste which the continual work of life entails. In the young, food serves a double purpose—it supplies material for growth, and it also affords substance from which the supply of force is derived. In the adult, whilst no doubt, to a certain extent, the food supplies actual loss of substance, it is more especially devoted to the performance of work, and of maintaining that equilibrium or balance between work and repair, which, as we have seen, constitutes health.

Viewed in this light, the first important rule for food-taking is founded on the plain fact, that in the food we must find the substances necessary for the repair of our bodies, and for the production of the energy through which work is performed. Food-substances in this light fall into two well-marked classes—namely, into *Nitrogenous* and *Non-nitrogenous* substances. Another classification of foods divides them into *organic* and *inorganic*, the former being derived from animals and plants—that is, from living beings—while the latter are derived from the world of non-living matter. Thus, animal and plant substances represent organic foods; while water and minerals, both of which are absolutely essential for the support of the body, represent inorganic food materials. It would appear that from living matter alone, do we obtain the materials for generating force. The inorganic water and minerals, however, appear to be absolutely necessary for the chemical alterations and changes which are continually taking place within the body.

Adopting the classification of foods into the *Nitrogenous* and *Non-nitrogenous* groups, we discover examples of the first class in such substances as *albumen*, seen familiarly in white of egg and other

substances; *gluten*, found in flour; *gelatin*, obtained from hoofs and horns; *legumin*, obtained from certain vegetables; *casein*, found in milk; and allied chemical substances. These substances possess a remarkable similarity or uniformity of composition. It would appear that in the process of digestion they are reduced to a nearly similar state, and on this account they can replace one another to a certain extent in the dietaries of mankind.

The nitrogenous foods have often been popularly termed ‘flesh-formers,’ and doubtless this name is well merited. For, as the result of experiment, it would seem that the chief duty performed by the nitrogenous parts of our food is that of building up and repairing the tissues of the body. They also produce heat, through being chemically changed in the blood, and thus aid in the production of force or energy. But it would also appear tolerably certain, that in a complex fashion the nitrogenous parts of our bodies assist or regulate in a very exact manner the oxidation or chemical combustion of the tissues.

It should be noted that nitrogenous foods are composed chemically of the four elements, carbon, hydrogen, oxygen, and nitrogen; the presence of the last element giving the characteristic name and chemical features to the group. Most of these foods in addition contain small proportions of sulphur and phosphorus.

An interesting advance in our knowledge of the part played by nitrogenous foods in the work of the body was made, when an idea of Liebig was overthrown by later experimentation. Liebig supposed that the nitrogenous foods required first to be actually converted into tissue—that is, into bodily substance—before their energy or work-producing power could be liberated. In this view, muscular force, through which we move, was believed to be dependent on the changes, destructive or otherwise, which take place in the muscles. The substance called *urea*, chiefly given off as a waste product by the kidneys and chemically representing nitrogenous waste, was in Liebig’s view regarded as representing the results of muscular force which had been exerted. But two scientists, Fick and Wislicenus of Zurich, proved, by a laborious series of personal experiments in mountain ascents, that a non-nitrogenous diet will maintain the body for a short time during the performance of severe work, no great increase in the amount of urea given off being noticed. The work in question was proved to have been performed on the carbon and hydrogen of the food consumed. These

experiments have led to the now accepted view, that a muscle, instead of losing substance during work and thus wasting, in reality consumes nitrogen, and grows. The exhaustion of the muscle is dependent not so much on chemical waste, as on the accumulation within it of the waste products of other foods. The muscle, in other words, is merely the agent whereby so much energy, derived from the food, is converted into actual and applied force. Did muscle really waste, as Liebig supposed, the heart's substance would be entirely consumed by its work of one week!

Such being the functions and nature of nitrogenous foods, we may now glance at the non-nitrogenous division. Four groups of foods are included in this latter class—namely (1) Starches and sugars, or ‘amyloids’ as they are often termed; (2) fats and oils; (3) minerals; (4) water. The *starches and sugars* include not merely starch and sugar, as ordinarily known, but various gums, and certain acids, such as lactic and acetic acids. Starch, as in bread, is a most important food. These foods appear to go directly to maintain animal heat, and to give energy, or the power of doing work, to the animal frame. The heat-producing powers of starches and sugars are certainly inferior to those of the fats and oils. But starches and sugars can be converted into fat within the system; and hence persons who suffer from a tendency to obesity are warned to exclude these foods from their dietaries. Starches and sugars likewise appear to assist in some measure the digestion of nitrogenous foods. That *fats and oils* are heat-producing foods is a fact taught us by the common experience of mankind that northern nations consume the greatest proportion of fat. The heat-producing powers of fat have been set down at two and a half times as great as those of starch and sugar; and there is no doubt that, in addition to assisting in the conversion of food into body substances, the fatty parts of our food also assist in the work of removing waste matters from the body. Fat, in addition, being chemically burned in the blood, gives rise to the force which we exert in ordinary muscular work.

The *mineral* parts of our food play an important part in the maintenance of the frame. We thus require iron for the blood, phosphorus for the brain and nerves, and lime for the bones; whilst a variety of other minerals is likewise found in the blood and other fluids of the frame. The uses of the mineral constituents of our body are still a matter of speculation. Small as may be the quantity of certain minerals required for the support of the body, serious

health-derangement may result when we are deprived of these substances. Thus, scurvy appears to be a disease associated with the want of the mineral potash in the blood; and the cure of this disease is therefore accomplished when we supply to the blood those mineral elements which have previously been deficient. Common salt, or chloride of sodium, as it is chemically termed, although not entering into the composition of the body, appears to form an important part of all the secretions; and there can be little doubt that this mineral aids the formation and chemical integrity of the gastric juice of the stomach.

Water forms the last item in the list of non-nitrogenous foods. Of all foods, perhaps, water is the most important, seeing that it is a substance which, in the absence of all other nourishment, can sustain life for a period numbering many days. Thus, whilst a man dies in from six to seven days when deprived of solid food and water, life may be prolonged to as many as sixty days on water alone. The high importance of water as a food is abundantly proved, when we discover that it constitutes about two-thirds of the weight of the body; that it enters into the composition of the brain to the extent of eighty per cent.; that the blood consists of nearly eighty per cent. of water; and that even bone contains ten per cent. of this fluid. Entering thus into the composition of every fluid and tissue of the body, and being perpetually given off from lungs, skin, and kidneys in the ordinary work of life, there is little wonder that water assumes the first place amongst foods. Regarding the uses of water as a food, we see that it dissolves and conveys other foods throughout the system; that it assists in removing waste products; and that it also takes a share in regulating the temperature of the body through its evaporation on the skin.

Having thus considered the chemistry of foods, we may now pass to discuss the natural rules which science describes for the health-regulation of life in the matter of diet. A primary rule for food-taking is that which shows that, for the due support of the body, we require a combination of nitrogenous and non-nitrogenous foods. This fact is proved by the consideration that milk, 'nature's own food,' on which the human being grows rapidly in early life, is a compound of both classes of foods. So also, in an egg, from which is formed an animal body, we find a combination of the two classes. Death results if we attempt to feed on either class alone; and as the body consists of both classes of substances, the justification for the combination of foods

is complete. Man can obtain the required combination of nitrogenous and non-nitrogenous foods from animals alone, from vegetables alone, or from animals and vegetables combined. The water, of course, which is an absolutely essential feature of all dietaries, is regarded as an additional item. In regulating the dietary of mankind, it is found that the food of nations is determined largely, or completely, by their situation on the earth's surface. Thus, the northern nations are largely animal feeders; whilst the southern peoples of the world are to a great extent vegetarians. Individual experience and taste produce amongst the units of a nation special proclivities in the way of diet. But we can readily see that mankind, with that elasticity of constitution and power to avail themselves of their surroundings, can adapt themselves to their environments, and become animal feeders, vegetable feeders, or subsist on a mixed dietary at will. This is the true solution of the vegetarian controversy. It is climate and race which determine the food of a nation. It is individual intelligence, liking, and constitution which determine variations and departures from the dietaries of the race.

The relations between food and work naturally present themselves as topics of the highest importance. In determining the standard of health, it is clear that from our food alone, we can obtain the energy or power of work required for the discharge of the duties of life. An interesting point therefore arises regarding the differences which are entailed by varying conditions and amounts of labour. Dr Letheby tells us that an adult man in *idleness* requires, to obtain from his food for the support of his body, 2.67 ounces of nitrogenous matter and 19.16 ounces of non-nitrogenous matter per day. If the individual is to participate in *ordinary labour*, the amount of nitrogenous matter obtained from his food must be increased to 4.56 ounces, while the non-nitrogenous must be represented by 29.24 ounces. In the case, lastly, of *active labour* the amount of food required must be increased to 5.81 ounces of nitrogenous, and 34.97 ounces of non-nitrogenous matter.

Dalton gives the following as the quantity of food, per day, required for the healthy man, taking free exercise in the open air: meat, sixteen ounces; bread, nineteen ounces; fat or butter, three and a half ounces; water, fifty-two fluid ounces. It ought to be borne in mind that these amounts of food represent the diet for a whole day compressed, so to speak, into a convenient and readily understood form. Another calculation, setting down

the daily amount of food required by an adult, at nitrogenous matter three hundred grains, and carbon at four thousand grains, shows that these amounts would be obtained from eighteen ounces of bread; one ounce of butter; four ounces of milk; two ounces of bacon; eight ounces of potatoes; six ounces of cabbage; three and a half ounces of cheese; one ounce of sugar; three-quarters of an ounce of salt; and water (alone, and in beverages) sixty-six and a quarter ounces—a total of no less than six pounds fourteen and a quarter ounces. Summing up the question of the amounts of food required by a healthy adult daily, and *excluding water in all forms as a matter of separate calculation*, it may be said that four and a half ounces of pure nitrogenous matter would be required in addition to three ounces of fatty food, fourteen ounces of starch or sugar, and one ounce of mineral matter. An ordinary adult consuming in twenty-four hours, food items equal to those contained in one pound of meat and two pounds of bread, may be regarded as consuming food of sufficient amount for ordinary work. When the work is increased, the diet must naturally be increased likewise. We find that persons in active employment require about a fifth part more nitrogenous food, and about twice the quantity of fat consumed by those engaged in light work; the sugars and starches remaining the same.

An interesting practical calculation has been made regarding the amounts of different foods required to perform a given and fixed piece of work. Taking the work performed by the German observers already named, as a standard, namely, that of raising a man's weight (one hundred and forty pounds) ten thousand feet high, it has been found that the amounts and cost of various foods required for the performance of this work is as follows: Bread, 2.345 pounds, cost 3½d.; oatmeal, 1.281 pounds, cost 3½d.; potatoes, 5.068 pounds, cost 5¼d.; beef-fat, 0.555 pounds, cost 5¼d.; cheese, 1.156 pounds, cost 11½d.; butter, 0.693 pounds, cost 1s. 0½d.; lean beef, 3.532 pounds, cost 3s. 6½d.; pale ale, nine bottles, cost 4s. 6d.

The proportion of the different food-elements in an ordinary dietary has been set down as follows: nitrogenous matter one, fats six, starches and sugars three; and these proportions appear to be represented with singular exactness in the ordinary dietaries which experience has recommended to mankind. Excess of food in the matter of nitrogenous elements tends to induce diseases of an inflammatory and gouty nature, and likewise leads to

fatty degeneration of the tissues. When, on the other hand, there exists lack of nitrogenous substances, the individual experiences weakness, want of muscular power, and general prostration. The healthy mean is that in which the proportions of nitrogenous and non-nitrogenous food are maintained as above indicated.

In the construction of dietaries, a few practical hints remain for notice. Thus, as regards sex, the dietaries of women are usually, in the case of the working-classes, estimated at one-tenth less than those of the opposite sex. Age has an important influence in determining the amount and quality of food. The growing body consumes more food, relatively to work and weight, than the adult, inasmuch as it requires material for new tissue. An infant under eight or nine months should receive no starch whatever in its dietary, because it is unable to digest that substance. Health is naturally a condition in which the question of foods assumes a high importance, and various dietaries, as is well known, are adapted for the cure of disease. The relation of food to work has already been alluded to, and statistics detailed; but it may be added that the brain-worker requires his food in a more readily digestible form, and also in smaller bulk and in more concentrated shape, than the muscle-worker or ordinary labourer. What has been said concerning foods will tend to show how wide is the field which the subject of nutrition occupies. It may only here be added, that the education of the individual in health laws and in the science of foods and food-taking, forms the only sure basis for the intelligent regulation of that all-important work—the nourishment and due support of the frame in relation to the work we perform and to every circumstance of life.

THE COMMON-SENSE OF SUPERSTITIONS.

OUT of a medley of magpies, May cats, broken looking-glasses, crickets, village cures, lucky days, and tumbles up-stairs, there dawns a hint towards the solving of a very puzzling problem. The problem is, not why these things are called lucky or unlucky, but how it is that multitudes grow up in every generation to believe the same absurdities, and that still in this world of common-sense such items of uncommon nonsense keep their character for 'coming true.' How is this, and where do the secret links exist between the sense and the nonsense? If any one takes the trouble to gather together about a hundred rustic superstitions and old beliefs of quackery, the reason of the character for 'coming true'—that is, the reason of the traditional hold upon the people—will presently begin to be plainly written across the whole medley, dawning by degrees, just as writing in acid might dawn upon an apparently blank missive held to the heat.

Most superstitions are signs of ill-luck. This in itself is a tell-tale fact. Unlucky omens are so numerous, that no believer could escape them for long; and in all likelihood he observes not only the unlucky signs, but his ill-luck following. The truth is, that the magpie on his path had no connection with his loss of money; and on his wedding-day, his bride's unlucky glance in the looking-glass after she was fully arrayed, had nothing to do with her discontent as a wife; nor need the servant who broke the looking-glass have cried, looking forward to seven years of ill-luck. In all three cases, as all the neighbours knew, the ill-luck came. But it came because of the prepossessed frame of mind that observed and discounted these signs. The superstitious character lacks those practical and courageous qualities which wrest luck from fortune and make the best of life. The omens of ill-luck have come to the fortunate as often; but they were never noticed, because they who were cheerily fighting the battle had better use for their time. At this moment, the present writer knows of no household more radiantly prosperous than one in which the largest looking-glass was broken a few days after a move to their newly-built home; and no marriage more replete with happiness than a Saturday marriage, though proverbially Saturday's marriage 'has no luck at all.' Of course, neither the prosperous

household nor the well-matched pair were of that languid and timid mind that takes nervous note of superstitions.

But, it may be objected, there are signs of good luck too, though not so many. Certainly; and there is no truth better known than that courage commands success, and such courage in exceptional cases may come from a very trivial encouragement. There is a country superstition that if a man sets off running and runs round in a circle, when he hears the cuckoo for the first time, he will never be out of work till spring comes again. But the man who valued steady work would exert himself in a more sensible direction than unproductive circle-running, and be safe from idle days. Again, if a tumble up-stairs is lucky, the predisposition to luck is in the person who will be active and quick enough to run up the staircase. Another good omen, the turning of a garment inside out in dressing, though it seems to tell of the slovenliness that will not succeed, has probably an origin that indicated something better; it is a country saying, and it might well refer to the hurry and awkwardness of rising without artificial light before day—a habit likely to help the farmer's household to good fortune. Or as proof of the real nature of many good signs which time has perverted into superstitions, can we doubt that the crickets which chirp round the hearth for luck were first noticed there because crickets, as a rule, only come to a warm and cosy fireside—the kind of hearth that marks a happy cottage home?

A simple grain of common-sense like this must have been the origin of many senseless observances. It was necessary to guard ladders from being knocked down, so superstition began to warn the passers-by: if the children went under the ladder, they would not grow; if girls went, they would have no chance of being married within the year; and if a man passed under, he would be hanged—in memory of the criminal's ladder under the gibbet.

To take another original grain of common-sense. Warnings against carelessness assumed the form of omens. To spill the salt was unfortunate; or in some country places, to spill new milk; or in parts of Southern Europe, to spill the oil. Leonardo da Vinci painted spilt salt near Judas in his famous 'Last Supper.' It is one of the most widespread of ill omens, though in different places there are shades of difference; for instance, in Holland it betokens a shipwreck.

Beside the superstitious disposition being what we may call an unlucky disposition, and beside the germ of encouragement that makes its own success out of some 'good signs,' and the atom of original prudence that still exists in some so-called bad omens, there are two other reasons why superstitions still keep hold of the people by a reputation for 'coming true.' These two reasons cover a great deal of ground in our theory of explanation. The first is the vague character of forecasts. For instance, we all know the rhymes about the luck of birthdays, which country-people of different shires repeat rather variously. One Scottish version is:

Monday's bairn is fair of face;
Tuesday's bairn is full of grace;
Wednesday's bairn is a child of woe;
Thursday's bairn has far to go;
Friday's bairn is loving and giving;
Saturday's bairn works hard for a living;
But the bairn that is born on the Sabbath day,
Is lively and bonnie, and wise and gay.

Contrast with this the English version:

Born of a Monday, fair in face;
Born of a Tuesday, full of God's grace;
Born of a Wednesday, merry and glad;
Born of a Thursday, sour and sad;
Born of a Friday, godly given;
Born of a Saturday, work for your living;
Born of a Sunday, never shall we want—
So there ends the week, and there's an end on't.

Any superstitious rustic who, from the page of the cottage Bible, dug out the deep secret of the day of his birth, would easily find the rhyme true of himself for any day of the week. Any country girl would trust it was true, if she was born on a Monday. And who that came on a Tuesday would confess himself graceless? But about Wednesday's bairn there seems to be a difference of opinion among the prophets: one rhyme predicts 'a child of woe;' the other says, 'merry and glad;' while a third, well known in Devonshire, says, 'sour and grum;' and thereby, from self-contradiction, the

old rhyme goes down like a house of cards. But all the rhymsters are agreed that Saturday's child works hard for his living—as no doubt the children of every other day of the week work too, in the sphere of labouring country-life in which these old sayings are known. And as variable as this forecast there are many others; for every firm believer in superstition has a secret satisfaction in proving it true; and which of us is there that could not read our life as the interpretation of any forecast, since we all can look at the bright or the dark side, having known alike the good and the evil days?

The other reason for the reputation for truth is, that, for credulous folk, unlucky omens are too terrible to be put to the test. If they were freely tried, they would be detected as a mental tyranny, a popular fraud; and in a few generations would be remembered by the rustic classes, only as the learned now remember the foolish excitement of their forefathers in science, seeking the Elixir of Life and the Philosopher's Stone. If dinner-parties of thirteen were to become the fashion, we should not see, as we often see now, the cautious arrangements of Christmas invitations, or even the timid compromise of bringing in a side-table to accommodate the thirteenth. But which of the credulous would dare to test these things? It reminds the writer of a doubt—still unsolved—whether the taste of parsley would cause a parrot to drop down dead. Parsley as a parrot-poison was heard of in childhood, not as a superstition, but as a physical fact. What if it were true? The *if* was too terrible. We had visions of our feathered gray 'Prince Charlie' seizing the green stuff in his hooked beak, and rolling off the perch in mortal agonies. So we disbelieved, but coward-like avoided the chance, just as all the world avoids thirteen at table.

As to superstitious cures, some of them contain slight elements of medicinal value; but most depend upon that influence upon the nerves which is well known to be capable of giving energy for a time and allaying pain. Some of the old cures were decidedly disagreeable and troublesome. The native of Devonshire who wanted to get rid of a wart was solemnly enjoined to steal a piece of meat, and after rubbing the wart with it, throw it over his left shoulder over a wall. The Hertfordshire villager, when afflicted with ague, might be cured if he would go to Berkhamstead, where oak-trees grew at the cross-roads; and after pegging himself by a lock of hair to the trunk of one of these trees, he was to give a vigorous jump, and rid himself at once of the ague and the tuft of hair. The loss of the hair was so painful, and the

loss of the ague so doubtful, that the Berkhamstead folks many years ago ceased to go to 'the cross-oaks.' The ague, the toothache, and dog-bites were the subject of many charms. In the former two maladies, a nervous impression might go far to cure; and in the last, a charm against hydrophobia would protect the simple believer from the great peril that is in a brooding fear of madness. The ludicrous cures were a legion in themselves. It seems heartlessly unkind to give a poor dog the measles; but many an old nurse took a lock of hair from the nape of the sick child's neck, made a sort of sandwich of it between bread-and-butter, and watched at the door to transfer, or fancy she had transferred, the measles to a stray dog—probably a stray dog, because only an ill-fed animal would take her bread. Equally unkind was it to strive to give our dumb friend the whooping-cough; but by the same process, with a bunch of hair and a piece of meat, the nurse could be guilty of that absurdity as well.

Have any of our readers ever encountered a toad with the whooping-cough? The Cheshire toads ought to be sometimes found crowing and whooping and in need of change of air; for the superstitious Cheshire woman whose child has the cough, knows that she has only to poke a toad's head into her child's mouth to transfer the whooping-cough to the toad. Query, Is the disease also transferred—and in that case, what are the alarming results—when the victim of whooping-cough gets rid of it by being passed nine times under and over a donkey? The cure for rickets is to pass the child under and over the donkey nine-times-nine turns. This was actually done in London as late as 1845; when a man and a woman, solemnly counting, passed the unfortunate child under and over the unsuspecting moke eighty-one times, in the midst of an admiring crowd. If there was one pass more or less, the charm would fail—a broad enough hint of the excuses that could be made when such cures as these were sought in vain. The eighty-one turns must have confused the counters' arithmetic, as no doubt the child had personal objections, and lifted its voice aloud; and sore must have been the trial even to the patience of a donkey.

So, to sum up, we would suggest that superstitions keep their false character for truth, firstly, because those who observe them therein prove their own leaning towards ill-luck; secondly, because forecasts are vague, and interpretations can be traced somehow in the chances of life; thirdly, because the penalty of ill omens is so dreaded, that the credulous shrink

from putting them to the test; fourthly, because there are nervous cures, and love-charms, and dreams, in which anxious consciousness points right—the wish being father to the thought; fifthly, victims of superstition are secretly pleased when (by chance) an unlucky omen comes true, and have a satisfaction even in relating their misfortunes; while, since no one tells of the cases that do *not* come true, every chance fulfilment is a new rivet in the chain that ought long ago to have fallen to pieces.

NOXIOUS MANUFACTURES.

THERE is just now a most wholesome activity in regard to the national health, and the public are peculiarly interested in the various details of our sanitary machinery. Of this, by no means the least important department is that instituted under the Alkali Works Regulation Act, 1881, or, in other words, the inspection of noxious works and factories. In connection with the pollution of rivers, this is an old grievance; but too little has hitherto been done to realise or to remedy the evil in its general effects upon the public health. So greatly, too, have works prejudicial to health increased of late years, that their inspection has been decided upon none too soon. Probably, it will never be known how far the death-rate has been influenced by this cause. It is, however, one of the unavoidable penalties of civilisation that we should live under unwholesome conditions of life.

A multitude of influences injurious to health spring into active existence with the development of commerce and the growth of luxury. Most of these are evident enough. All the elements, indeed, are equally guilty. The earth, air, fire, and water, are allied against civilised humanity; and modern science is constantly bringing to light disagreeable facts in this connection. We have long lived in the comfortable belief that Mother Earth was the great purifier. The reverse is, it seems, nearer the truth. Years after the germs of infection have been consigned to the ground, they have been disinterred, and found to be not a whit diminished in virulence. Archæologists should, we are told, beware of handling newly found relics, lest, perchance, they should contract some archaic disease. Even mummies, it appears, in spite of their venerable respectability, are objects of legitimate suspicion! Fire, too, has a dreary catalogue of sins to answer for. It not only robs us of much of the oxygen, of which those of us who live in the towns have so scanty a supply, but it gives us in exchange unconsumed carbon in quantities which fill the air with smut. In smoke alone it furnishes us with food for reflection—and digestion—and probably will continue to do so for some time to come.

Again, water is the most insidious enemy of all. The most indispensable of the elements—and we are reminded of our obligations to it pretty frequently—it is credited with doing the greatest harm. In league with unnatural substances, it has developed such an affinity for noxious matter that it appears that nothing short of boiling can possibly enable us to drink it with any security. To most people, cold boiled water will not seem a very attractive beverage, but it has the advantage of being in many ways a safe one.

The air, too, is anything but true to the trust committed to her charge. We have long confidently believed in her good-will. Our sewers, drains, and chimneys discharge their pestilent exhalations into the air; but instead of carrying these away into space, she receives them only to bestow them upon us again.

The outlook is indeed gloomy, and unless we make some progress in sanitary science, it is not a little difficult to see how we are to continue to support the burden of civilised existence.

In this connection, it is reassuring to know that something is being done to lessen these ominously numerous artificial dangers. The works which come within the scope of the Alkali, &c. Works Act, 1881, are very injurious to life. The manufacture of alkalies, acids, chemical manures, salt, and cement, alike involve processes prejudicial to health. More than one thousand of these were visited by the inspectors, appointed in pursuance of the above Act, during the year 1882; and it is interesting to know that some intelligent means are being devised whereby the offensive character of these manufactures may be diminished. To take a single cause of mischief. The manufacture of alkalies and acids has long been conducted in such a way that the proportion of noxious matter which was allowed to escape into the chimney, or atmosphere, often reached from twenty to forty grains per cubic foot of air, twenty being a not uncommon amount. The maximum amount which might be allowed to escape with impunity has been estimated at four grains per cubic foot; and it is a very important feature of the Act that it has been instrumental in reducing this very considerably. In the alkali works proper the escape has been brought down to two grains, while in some cases it is under one. The sulphuric acid works alone are now conspicuous for their failings in this important respect, the average escape in those examined during the year being 5.5. Again, chemical manure-works have

long been a pregnant source of annoyance to the inhabitants of the neighbourhood in which these are carried on.

It is, curiously enough, the smaller establishments of the kind which are the most harmful. The larger works have long employed the most complete processes, because the escape of effluvia would otherwise have been so great, that it would have speedily aroused hostile action on the part of the public. The imposition of preventive measures in the case of the smaller works—in many of which no precautions whatever have hitherto been adopted—is attended with some difficulty, since it involves an expenditure which would in some cases be almost prohibitive. It appears, indeed, that no maximum of escape can be fixed in works of this kind, and all that remains to be done is to render it compulsory that processes should be adopted for washing out such gases as are soluble, and for burning those which are more susceptible to such a method of treatment. Since such pernicious agents as fluorine compounds escape during the action of sulphuric acid upon phosphates, the question is one of some urgency. Again, another cause of complaint is the escape of sulphuretted hydrogen during the process of making sulphate of ammonia. In the larger gas-liquor-works the gas is burned, and converted into sulphuric acid in lead chambers; while in others it is passed through oxide of iron; and both these methods are perfectly satisfactory when properly carried out. Again, the discharge of sulphurous or muriatic gases evolved in extracting salt from brine is an evil which has remained unremedied almost down to the present time. Not the least curious feature of this question, too, is the fact that many of the products of distillation are so valuable that it is more than mere neglect to throw them away in the form of noxious gases. It is unnecessary to describe here the state of the salt districts. They might serve as a type of the abomination of desolation. The combined effect of the gases and the soot, which pours forth in prodigious volumes and from the chimneys of nearly a hundred salt-works in Cheshire alone, is most deplorable.

The only possible conclusion from this Report is that we are still far behindhand in these matters. We have, for instance, long continued to burn coal on the same principle, and are very slow to believe in any of the new methods which have been and are continually being introduced. Yet not only is black smoke very much more injurious to animal and vegetable life than when it has been rendered colourless by burning, but it is peculiarly

wasteful. It has long been known that many valuable commodities could be obtained from coal; and but too little progress has hitherto been made in this direction. It is, then, all the more interesting to know that in some works in the north of England the gases from the blast furnaces have been cooled and washed, and ammoniacal salts obtained in such quantities as to make the process economical; while by the 'Young and Beilby' process it is contended that not only can the fuel be consumed for nothing, but that there will be several shillings a ton profit.

So far as manufactures are concerned, there certainly seems to be no valid reason why the rule that they must consume their own smoke should not be much more freely enforced. In the case of the alkali trades, which have long been in a very bad state, it is, of course, an unfortunate time to suggest the necessity for the outlay of more capital in improved works. But the exigencies of the public health are paramount, and needlessly offensive processes cannot be tolerated much longer. Such a case as that reported from Widnes, where waste heaps of offensive matter, consisting chiefly of sulphur and lime, are allowed to accumulate, although the sulphur could be extracted at a profit, and so prevented from poisoning the streams for miles around, is certainly difficult to explain. The drainage from these heaps alone is estimated as carrying away twelve tons or seventy pounds-worth of sulphur a day. But perhaps as soon as some satisfactory system for eliminating the sulphur has been hit upon, this will be remedied. We have certainly much yet to learn in sanitary science. The old theories are one by one being exploded, and it will no longer do for us to poison the air we breathe, under the pleasing impression that its purifying properties are inexhaustible. Civilisation has made such strides that she has succeeded in overturning the equilibrium of nature. The equilibrium must be restored.

TRIMMING THE FEET OF ELEPHANTS.

The feet of elephants kept for show purposes are trimmed two or three times a year. The sole of an elephant's foot is heavily covered with a thick horny substance of material similar to the three toe-nails on each foot; and as it grows thicker and thicker, it tends to contract and crack, often laming the animal. Barnum the American showman recently subjected his elephants to the trimming process at one of the towns where he was exhibiting. With a knife about two feet long, great pieces of horn, six inches by four, and a quarter of an inch thick, were shaved off. Often pieces of glass, wire, nails, and other things are found imbedded in the foot, which have been picked up during street parades. Sometimes these irritating morsels work up into the leg and produce a festering sore. A large nail was found imbedded in the foot of one of the elephants, which had to be extracted with a pair of pincers, and the wound syringed with warm water. During the operation, the huge creature appeared to suffer great pain, but seemed to know that it would afterwards obtain relief, and therefore bore it patiently, and trumpeted its pleasure at the close. Three times around an elephant's front-hoof is said to be his exact height.

SONNETS OF PRAISE.

THE VALES.

The nestling vales lie sheltered from rough winds,
As little babes in tender keeping grow,
Some narrow gorge each flowery limit binds;
Thus we from childish eyes hide elder woe.
The vales are thick with corn, with plenty shine;
Thus should the children smile in sunny glee,
For One hath blessed them with a love divine,
The untried pilgrims of life's stormy sea.
Though rough winds cannot enter, gentle rain
Refreshes the green vale, till springs arise,
Their source the snow-clad hills; so age should gain,
By gentle teaching childhood's eager eyes.
Rain fills the pools, the thirsty vale is blest;
Thus should the children thrive, by love caressed.

THE MOUNTAINS.

The lofty mountains with their snowy crests,
God's ensigns, praise their Lord throughout the land;
Their heights, which few can reach, in human breasts
Inspiring awe, yet quake beneath His hand.
Oft 'twixt their summits and the lower earth,
The wreathing cloud-mists roll, alone they dwell
As sight-dimmed age. Our cries of pain or mirth
Molest them not; thus age with deadening spell
Benumbs our ears, yet near each lonely peak
Sing mountain birds, sunbeams each summit crown.
From highest heaven thus God's saints may seek
Refuge in thoughts divine, though long years drown
Earth's sounds; on mountain crest reposed the Ark,

Our home above shines clear, as earth grows dark.

M. P.

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[Transcriber's note—the following changes have been made to this text.

Page 236: missing word “pounds” inserted—“3.532 pounds”.]

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